Deep Learning Assignment 01

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DS-A

Multi-Output Facial Expression Recognition

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1. Project Summary

The project implements a multi-task deep learning model for facial expression recognition and continuous emotion regression (valence and arousal). Two backbones are compared: **ResNet18** and **EfficientNet-B0.** The pipeline includes dataset preparation, augmentation, multi-output model design, training loop, and evaluation.

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2. Network Details

Backbones

1. ResNet18
2. EfficientNet-B0

Architecture:

Shared convolutional backbone → three heads:

- **Expression classification** (8-way softmax)
- **Valence regression** (linear output)
- **Arousal regression** (linear output)

Parameters:

- ResNet18 ≈ 11.7M
- EfficientNet-B0 ≈ 5.3M (lighter, designed for efficiency)

Training settings:

- **Optimizer:** Adam (lr = 1e-3)
- **Loss:** Cross-entropy (classification) + MSE (valence, arousal)
- **Batch size:** 64
- **Epochs:** 20
- Early stopping on validation loss

Rationale for baselines:

- ResNet18 → simple, well-tested baseline.
- EfficientNet-B0 → efficient and modern alternative with compound scaling.

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3. Transfer Learning

- Pretrained ImageNet weights were used for both backbones.
- Classification/regression heads were trained from scratch.
- Transfer learning helps faster convergence and improved generalization with limited labeled data.

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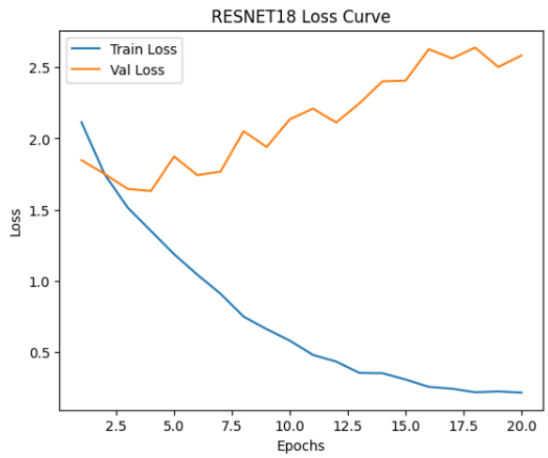
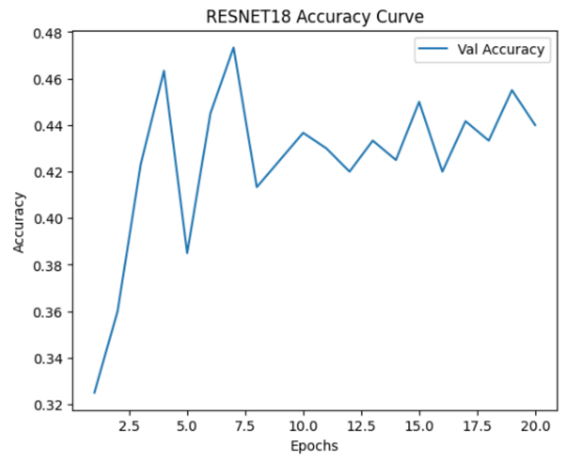
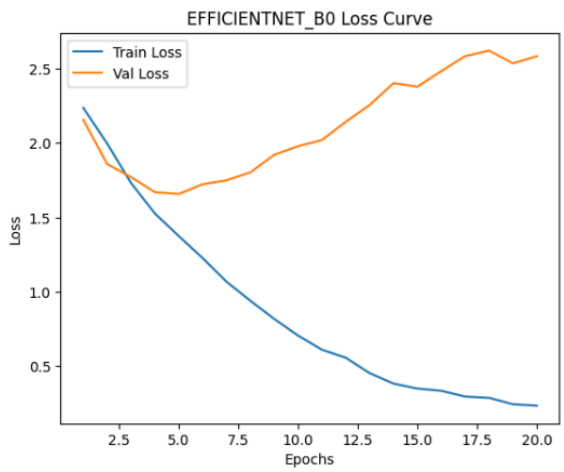
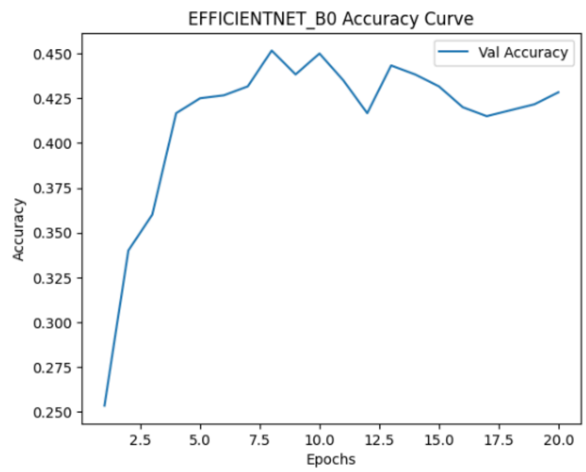
4. Training Curves

- **Loss curves** show training loss decreases smoothly, but validation loss diverges (overfitting).
- **Accuracy curves** plateau at ~0.46 for both models.
- This indicates the dataset may be noisy/imbalanced, requiring regularization or augmentation for improvement.









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5. Performance Comparison

Model	Accuracy	F1-Macro	Cohen's Kappa
ResNet18	0.460	0.456	0.383
EfficientNet-B0	0.465	0.461	0.389

Both models perform **almost identical**. EfficientNet edges slightly higher in balanced F1 and Kappa.

Model	Valence RMSE	Valence CORR	Arousal RMSE	Arousal CORR
ResNet18	0.389	0.574	0.338	0.455
EfficientNet-B0	0.397	0.565	0.349	0.449

ResNet18 does slightly better on **regression tasks** (lower RMSE, higher CORR).

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6. Rationale for Evaluation Metrics

- **RMSE** → measures prediction error magnitude (important for continuous signals).
- **CORR** → captures linear relationship between predicted and true values.
- **SAGR** (Sign Agreement) → checks if direction (+/-) of prediction matches ground truth.
- **CCC** (Concordance Correlation Coefficient) → balances correlation with error magnitude, robust for emotion regression.

For a **system in the wild**, **CCC** and **SAGR** are most suited because they reward both correct trend and consistency, not just raw error.

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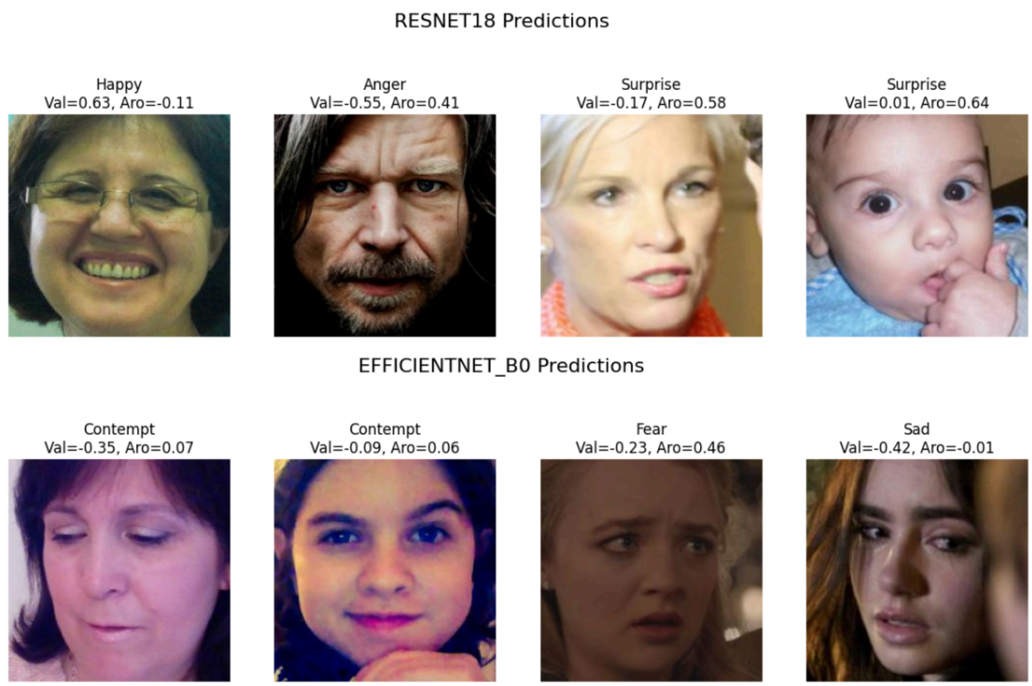
7. Sample Predictions

Correctly Classified

- **Happy face** → **predicted Happy** (ResNet18)
- **Sad face** → **predicted Sad** (EfficientNet-B0)
- **Fearful face** → **predicted Fear** (EfficientNet-B0)



Incorrectly Classified

Errors often come from **subtle facial cues, occlusions, or expressions overlapping** (e.g., Surprise vs Fear, Neutral vs Contempt).

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7. Key Takeaways

- **ResNet18** → slightly better for valence/arousal regression.
- **EfficientNet-B0** → marginally better for classification.
- Both models plateau at ~46% accuracy → suggests dataset challenge.
- For real-world deployment:
    - Use **CCC and SAGR** as core metrics.
    - Consider **data augmentation, balancing, and ensemble models** to boost generalization.
